

CANDIDATE CHALLENGE

Predict & Create Next-Season Styles

Use retail data to find the next winners, then turn them into new product ideas.

TIMEBOX	SUBMISSION	DISCUSSION
6–8 hours	Within 5 calendar days	Short walkthrough + Q&A

The outcome One image showing three new product concepts for next season, each inspired by a style your model predicts will perform strongly.

The challenge

Imagine you are working with a fashion retailer that wants to make better decisions about what to create next season. You have historical transactions, product information, and catalogue images.

Use the data to predict the three styles most likely to perform strongly in a future period. Then use generative AI to create one new product concept inspired by each of those styles.

The new concepts should keep the qualities that may have helped the original styles succeed, while still feeling like genuinely new products rather than copies.

The dataset

Use the H&M Personalized Fashion Recommendations dataset. It includes customer transactions, detailed article information, and product images linked through article IDs.

H&M Personalized Fashion Recommendations: [Open dataset](#) — Kaggle. You will need transactions_train.csv, articles.csv, and the matching product images.

The transaction data can be used to understand how products performed over time. The article data describes details such as product type, garment group, colour, department, and product description. The catalogue images provide the visual reference for the generation stage.

A note about stock This dataset does not include reliable stock-on-hand or availability history. Work with the transactions provided, explain how this limitation affects your prediction, and do not invent missing inventory data.

What we would like you to build

1. Predict the winners

Build a model that predicts which styles are most likely to perform strongly in the next period. You can decide how to define a style, what strong performance means, and how far ahead to predict. We are interested in the reasoning behind those choices as much as the model itself.

2. Create the next-season concepts

Take the top three predicted styles and use their images and product details as references. Generate one new look-alike concept for each winner. The result should carry forward the strongest visual ideas while introducing meaningful design changes.

3. Show an agent-based approach

Build the workflow using an orchestrating agent, specialist sub-agents, MCP-compatible tools, and at least one reusable skill. How you divide the work is up to you. We want to see a sensible design and a working demonstration, not a particular framework.

What to share

Please share:

- One final image showing the three generated next-season concepts.
- Your working code or prototype.
- A short explanation of your approach, key choices, results, and limitations.
- Enough evidence to show how the selected styles led to the generated concepts.

Bonus

Seasonal view Bonus points if you split performance by season and show how the top styles or generated concepts change for a chosen season.

There is no single correct solution. Keep the scope practical and be ready to talk us through the choices you made.